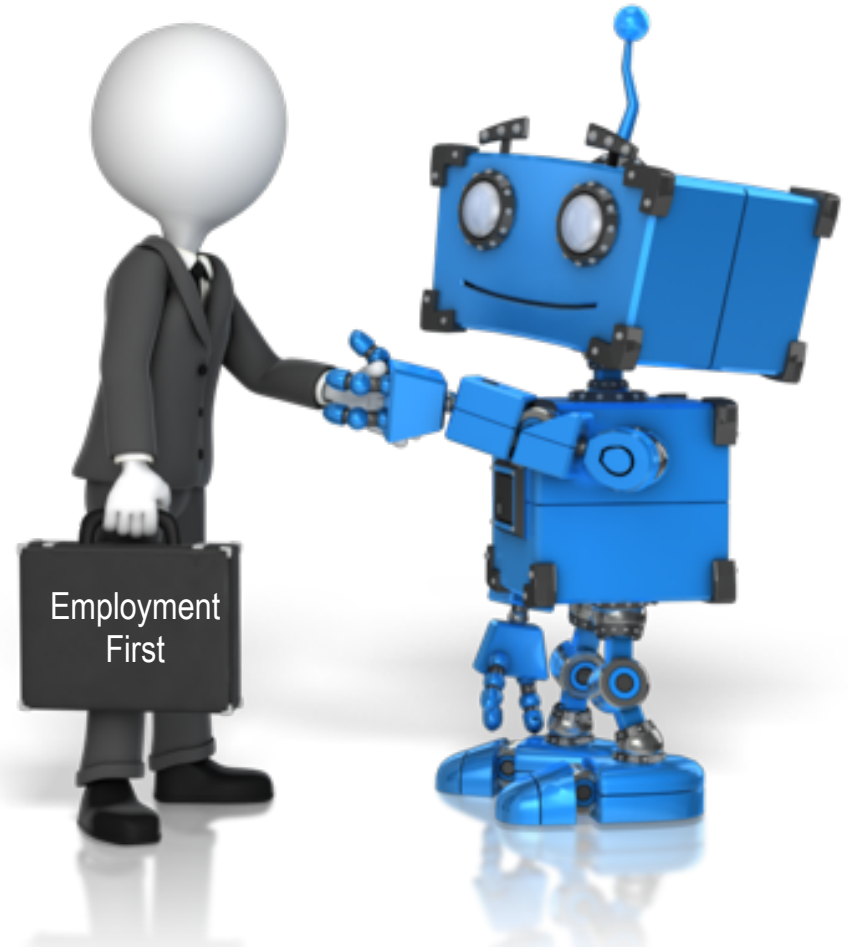


# Employment First Meets Technology First: *It's More Than Just The Device*



**David Hoff & Jessi Keenoy**  
Institute for Community Inclusion  
*UMass Boston*



“Technology and information access today is no longer a luxury.

It is essential for people with cognitive disabilities to have the same access as their peers without such disabilities in order to fully participate in the worlds of today and tomorrow.”



Coleman Institute for Cognitive Disabilities  
UNIVERSITY OF COLORADO

# The Rights of People with Cognitive Disabilities to Technology and Information Access

## *Whereas*

- Twenty-eight million United States citizens have cognitive disabilities such as intellectual disability; severe, persistent mental illness; brain injury; stroke; and neurodegenerative disorders such as Alzheimer's disease;
- People with cognitive disabilities are entitled to inclusion in our democratic society under federal laws such as the Americans with Disabilities Act (ADA), the Developmental Disabilities Assistance and Bill of Rights Act (DD Act), the Individuals with Disabilities Education Act (IDEA), Section 504 of the Rehabilitation Act, and under state and local laws;
- The disruptive convergence of computing and communication technologies has substantially altered how people acquire, utilize, and disseminate knowledge and information;
- Access to comprehensible information and usable communication technologies is necessary for all people in our society, particularly for people with cognitive disabilities, to promote self-determination and to engage meaningfully in major aspects of life such as education, health promotion, employment, recreation, and civic participation;
- The vast majority of people with cognitive disabilities have limited or no access to comprehensible information and usable communication technologies;
- People with cognitive disabilities must have access to commercially available devices and software that incorporate principles of universal design such as flexibility and ease of use for all;
- Technology and information access by people with cognitive disabilities must be guided by standards and best-practices, such as personalization and compatibility across devices and platforms, and through the application of innovations including automated and predictive technologies;
- Security and privacy must be assured and managed to protect civil rights and personal dignity of people with cognitive disabilities;
- Enhanced public and private funding is urgently required to allow people with cognitive disabilities to utilize technology and access information as a natural consequence of their rights to inclusion in our society;
- Ensuring access to technology and information for the 28 million people with cognitive disabilities in the United States will create new markets and employment opportunities; decrease dependency on public services; reduce healthcare costs; and improve the independence, productivity, and quality of life of people with cognitive disabilities.

## *Therefore*

We hereby affirm our commitment to equal rights of people with cognitive disabilities to technology and information access and we call for implementation of these rights with deliberate speed.

*View endorsers of this document and join us at: [colemainstitute.org/declaration](http://colemainstitute.org/declaration)*

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# TECHNOLOGY FIRST





- Technology considered as source of support for all individuals.
- Technology preferred over staff interventions.
- Informed choice and self-determination as drivers.
- Ability to become competent in technology is presumed.
- Development of competency in technology begins early and is ongoing.
- Consideration of technology is incorporated within all components of services.





## *Marriage of Employment First MO & Technology First MO*



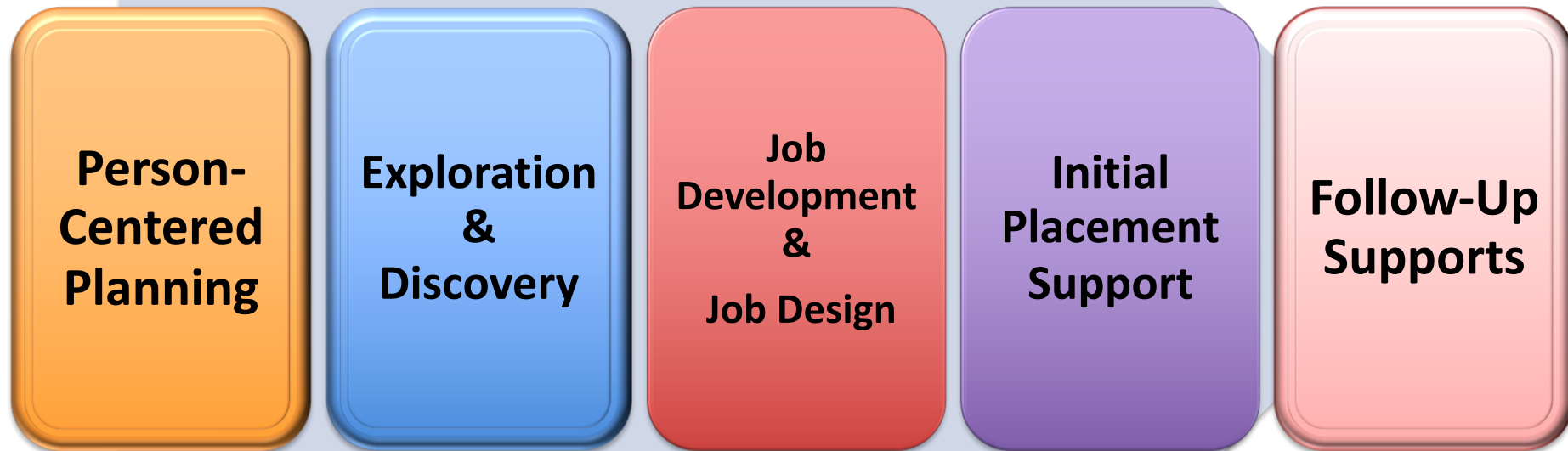
- Employment First MO asked to support Technology First MO
- Community of Practice, facilitated by ICI
  - **Members:** 3 employment providers, MO AT project, DD staff, DD council
  - **Meets monthly:** sharing technology ideas & implementation, capacity-building strategies
  - **Outputs:** Developed framework, written products

# SCOPE OF FOCUS

- Readily Available
- Embedded Apps & Tools
- Accessibility Features
- Apps
  - Video modeling
  - Communication
  - Task sequencing
  - Focus & sustained effort
  - Mindfulness
  - Scheduling



# The Employment Process

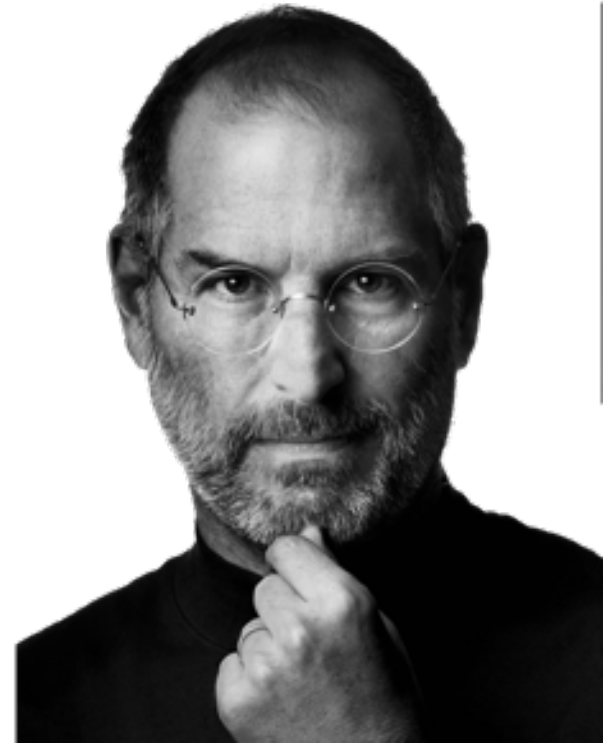


*Please share in the chat your ideas for integrating technology within the employment process.*



# STEVE JOBS

“I don’t focus on the technology.  
I focus on what’s in it  
for the customer.  
I want to give the  
customer an incredible  
experience.”



But know  
where to  
find me!



**Tech Expert**

# APPROACH TO TECHNOLOGY



Empower staff

Build on existing strengths & skills

- **Person-centered** and **individualized**
- **Solution-focused**
- **Self-determination & skill development**
- **Not all or nothing** – in combination with other supports

# TECHNOLOGY FRAMEWORK

**Individual**

**Environment**

**Implement  
&  
Evaluate**

**Narrow the Gap**

**Gap & Barrier  
Analysis**

# Individual



*Feel free to share examples of these factors in the chat.*

- Challenging tasks
- Strengths
- Compensatory strategies
- Self-maintenance strategies
- Learning preference or style
- Progressive or stable nature
- Natural motivators
- ***Access & experience with tech***
- ***Technical skill level***
- ***Technology preferences***
- ***Current supports available***
- ***Accessibility needs***



## Technology Availability and Skills Worksheet



Name:

Staff member:

Date:

### Sources of information:

**Instructions:** Record information on the form below regarding technology personally available to the individual (e.g., at home, or personal smart phone) and skills in using technology. The individual should be a primary source of information, with that information supplemented by others as appropriate (family, housemates, staff, etc.). For each item, indicate whether it is available by checking the box underneath. For those items that are available, indicate whether it is connected to the internet, the individual's ability to use the item, and whether the individual needs support to use the item. Add comments in the comment field. In conjunction with completing the form, answer the questions at the bottom.

| DEVICES               |                          |                                     |                          |                          |                          |                             | Comments |
|-----------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-----------------------------|----------|
| Computer              | Available<br>(check)     | Connected<br>to Internet<br>(check) | Ability to Use           |                          |                          | Needs<br>Support<br>(check) |          |
|                       |                          |                                     | Yes                      | Limited                  | No                       |                             |          |
| PC Desktop            | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| MAC Desktop           | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| PC Laptop             | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| MAC Laptop            | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| Chromebook            | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
|                       | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| Tablet                | Available<br>(check)     | Connected<br>to Internet<br>(check) | Ability to Use           |                          |                          | Needs<br>Support<br>(check) | Comments |
|                       |                          |                                     | Yes                      | Limited                  | No                       |                             |          |
| iOS-Apple (iPad)      | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| Android-Google        | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| Windows-<br>Microsoft | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| Blackberry-RIM        | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
|                       | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| Smartphone            | Available<br>(check)     | Connected<br>to Internet<br>(check) | Ability to Use           |                          |                          | Needs<br>Support<br>(check) | Comments |
|                       |                          |                                     | Yes                      | Limited                  | No                       |                             |          |
| iPhone                | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| Android               | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
| Windows Phone         | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |
|                       | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    |          |





# Environment



*Feel free to share examples of these factors in the chat.*

- In what environment is the barrier occurring?
- Resources available to everyone in the environment
- Policies & attitudes related to technology
- Required tasks
- Safety, security, or environmental concerns that may impact the use of tech
- Available supports
- Environmental infrastructure

# Gap & Barrier Analysis



- Task or tasks that are not being completed to expectations
- Environmental barrier
- Individual barrier or skill gap
- Interaction between environment and individual barrier
- What strategies have been tried in this environment and how effective have they been?

*Feel free to share examples of these factors in the chat.*

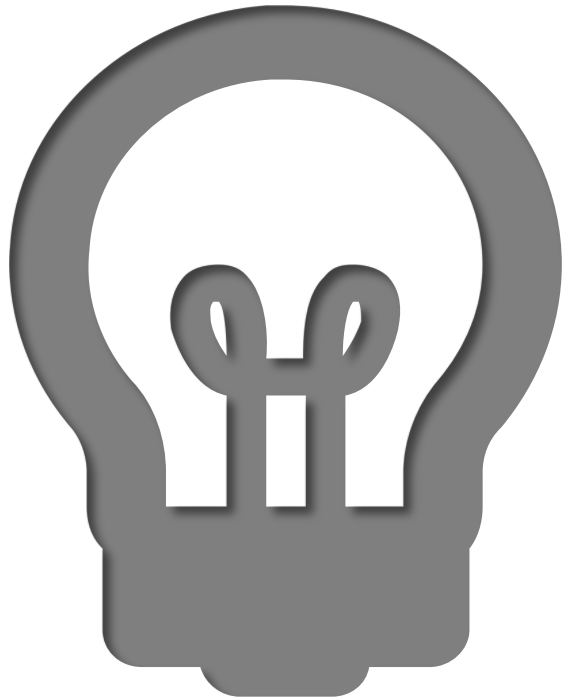
# Narrow the Gap



*Feel free to share examples  
of these factors in the chat.*

- Necessary skills & training needs
- Costs associated with tech
- Could the tech be distracting or isolating?
- Safety or security concerns
- Supports available to assist
- Is the tech adaptable?
- Who will own the technology?
- How will the tech be transported?
- Availability of a demo or trial
- Accessibility features
- Is tech associated with a request for an accommodation?
- Age appropriateness

# Implement & Evaluate



*Feel free to share examples of these factors in the chat.*

## Follow up considerations:

- Is the tech working?
- Person is able to access & operate tech
- Technology works in the environment
- Are task expectations being met?
- Is the tech creating distractions or barriers?
- Feedback from the individual, employer, & supports
- Necessary environmental changes
- Further training or supervision needs
- Unintended impacts
- Consider task analysis to assess data

# STEP BY STEP APPROACH TO APPLYING THE FRAMEWORK

## TECHNOLOGY ASSESSMENT

**Instructions:** This tool is intended to assess whether a technology app may be an effective strategy to use to increase an individual's independence at work or in the community. When used appropriately, the use of technology can decrease the time it takes to learn new tasks and can be effective in overcoming barriers. However, apps are not always the best option and decision making should be an individualized process that honors individual preferences. Often, a low or "no" tech solution is the best option to try first, before exploring the use of a technology app.

In order to assess whether the use of an app will be an appropriate strategy, many factors need to be taken into consideration including individual preferences and experiences, environmental characteristics and workplace policies, a comprehensive understanding of the barrier or gap, and a detailed analysis of the considerations associated with the potential solution. Once a solution has been identified and implemented, it is important to continuously re-assess the efficacy of the app in order to determine if it is still working as intended. As individual skills, work responsibilities, work environments, and technological advances are constantly changing, it is important to evaluate whether the app in use is the best fit or whether another solution may produce more desirable outcomes.

### Individual:

Age of Individual:

Employment Service Provider:

Current/Prospective Job:

Problem/barrier to be addressed:

Dates Completed:

Sources of Information used to complete assessment:

| Assessing the Individual                                                              | Response        |                      |
|---------------------------------------------------------------------------------------|-----------------|----------------------|
| What does the individual do really well related to his/her job or community activity? |                 |                      |
| What are some other strengths of this individual, not related to his/her job?         |                 |                      |
| What tasks does this individual generally struggle with?                              |                 |                      |
| What specific factors contribute to these concerns?                                   |                 |                      |
| How does this individual typically learn new tasks?                                   | Choose an item: | Additional Comments: |

- Piloting an assessment tool
- Domains can be used together or individually
- Variety of sources and methods to gather information
- Focus on one barrier at a time
- Honor individual choice & preferences
- Involve support team



- Incorporate into staff training, job description, competency requirements
- Leadership buy-in & support
- Engagement of individuals at **every step**
- Communities of practice
- Get help from the experts



- Contact information for state AT programs:  
[www.at3center.net](http://www.at3center.net)



## OTHER RESOURCES



*Feel free to share additional  
technology resources in the chat*



# ICI Publications

<https://covid19.communityinclusion.org>



## TOOLS FOR INCLUSION

ISSUE NO. 40 • 2020

**COVID-19**  
Day and  
Employment  
Services

### Using Technology for Remote Support, Self-Management, and Success in Employment and the Community

By David Hoff

Due to the COVID-19 pandemic, day and employment service providers have had to quickly determine how best to provide supports for people they serve. For many this means transitioning to services and supports that can be provided remotely. This brief is one in a series developed to help community providers continue to provide services to those who are currently working, seeking employment, or engaged in community-based day supports.

During this time, we are all learning how to best support the people we serve. We welcome your feedback and ideas as we all work together to provide services that help people to continue to work towards their goals while staying safe and healthy.

ICI COVID-19 resources:  
[covid19.communityinclusion.org](https://covid19.communityinclusion.org)

|                                                                         |   |
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| Technology Checklist.....                                               | 3 |
| Using Technology to Job Coach Remotely.....                             | 4 |
| Paying for Technology.....                                              | 5 |
| Identifying Appropriate Technology.....                                 | 6 |
| Using Technology for Task Management and Effective Job Performance..... | 7 |
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Assistive technology devices are nothing new. They were first defined in the Technology Related Assistance to Individuals with Disabilities Act of 1988 as “any item, piece of equipment, or product system, whether acquired commercially off the shelf, modified, or customized, that is used to increase, maintain, or improve functional capabilities of individuals with disabilities.” Assistive technology consists of a broad range of both low tech and high-tech devices, ranging from binder clips and Velcro, to highly customized computer workstations.

While assistive technology is not new, what has changed is high-tech devices for a broad range of purposes are now easily available and used almost universally. As a result, there is huge potential to make fundamental changes in the way services and supports are provided to individuals with disabilities. This publication discusses how readily available pieces of technology (smartphones, tablets, smart watches and similar devices) and their various features and capabilities, can and are being used as assistive technology to support individuals in employment and other community activities in ways that maximize success and independence, and reduce reliance on job coaches and other external paid supports.

Along with general guidelines on use and funding of technology, technology is examined from two specific perspectives:

- Providing job coaching in a way that does not require the physical presence of the job coach at the employment site.
- Using readily available technology to increase self-management of tasks and job performance.

While this publication focuses primarily on employment, these strategies can also be used to support individuals in community activities such as volunteering and recreation, and at home.

INSTITUTE FOR COMMUNITY INCLUSION  
University of Massachusetts Boston



## TOOLS FOR INCLUSION

ISSUE NO. 39 • 2020

**COVID-19**  
Day and  
Employment  
Services

### Easy to Access Assistive Technology and Apps for Individual Success

By Mary DiBlase and David Hoff

Due to the COVID-19 pandemic, day and employment service providers have had to quickly determine how best to provide supports for people they serve. For many this means transitioning to services and supports that can be provided remotely. This brief is one in a series developed to help community providers continue to provide services to those who are currently working, seeking employment, or engaged in community-based day supports.

During this time, we are all learning how to best support the people we serve. We welcome your feedback and ideas as we all work together to provide services that help people to continue to work towards their goals while staying safe and healthy.

ICI COVID-19 resources: [covid19.communityinclusion.org](https://covid19.communityinclusion.org)

The last several years have seen significant growth in the use of assistive technology (AT) and apps for supporting individuals with disabilities in employment, community activities, and related areas. Personal devices such as smartphones can serve as AT tools to support individuals with establishing and supporting personal goals as well as with employment supports.

This brief focuses on the various AT options that can be accessed through smartphones and tablets, often for free, to support employment, personal, and independent-living goals. Much of the technology discussed in this publication can also be accessed via desktop and laptop computers.

The COVID-19 pandemic presents an opportunity to spend time with individuals exploring and learning to use different AT and apps that can be used now and in the future. This includes technology and apps that can assist them to be more self-sufficient on the job and in the community. In addition, there are apps that can help with exploring employment options and conducting a successful job search, and that support individuals in community participation. There are also apps that individuals can use now to support personal wellness, and deal with stress and anxiety, as well as learn new skills such as money management.

For individuals who are working during the pandemic, at a time when in-person supports from a job coach may be limited or not available, AT and apps can increase their ability to self-manage on the job. Through use of technology, job coaching services can also be provided remotely. (See the other publication in this series [Using Technology for Remote Support, Self-Management, and Success in Employment and the Community](#), for how the types of assistive technology and apps discussed in this publication can be utilized to enhance success in employment and the community, and provide supports remotely.)

### Identifying Appropriate Technology

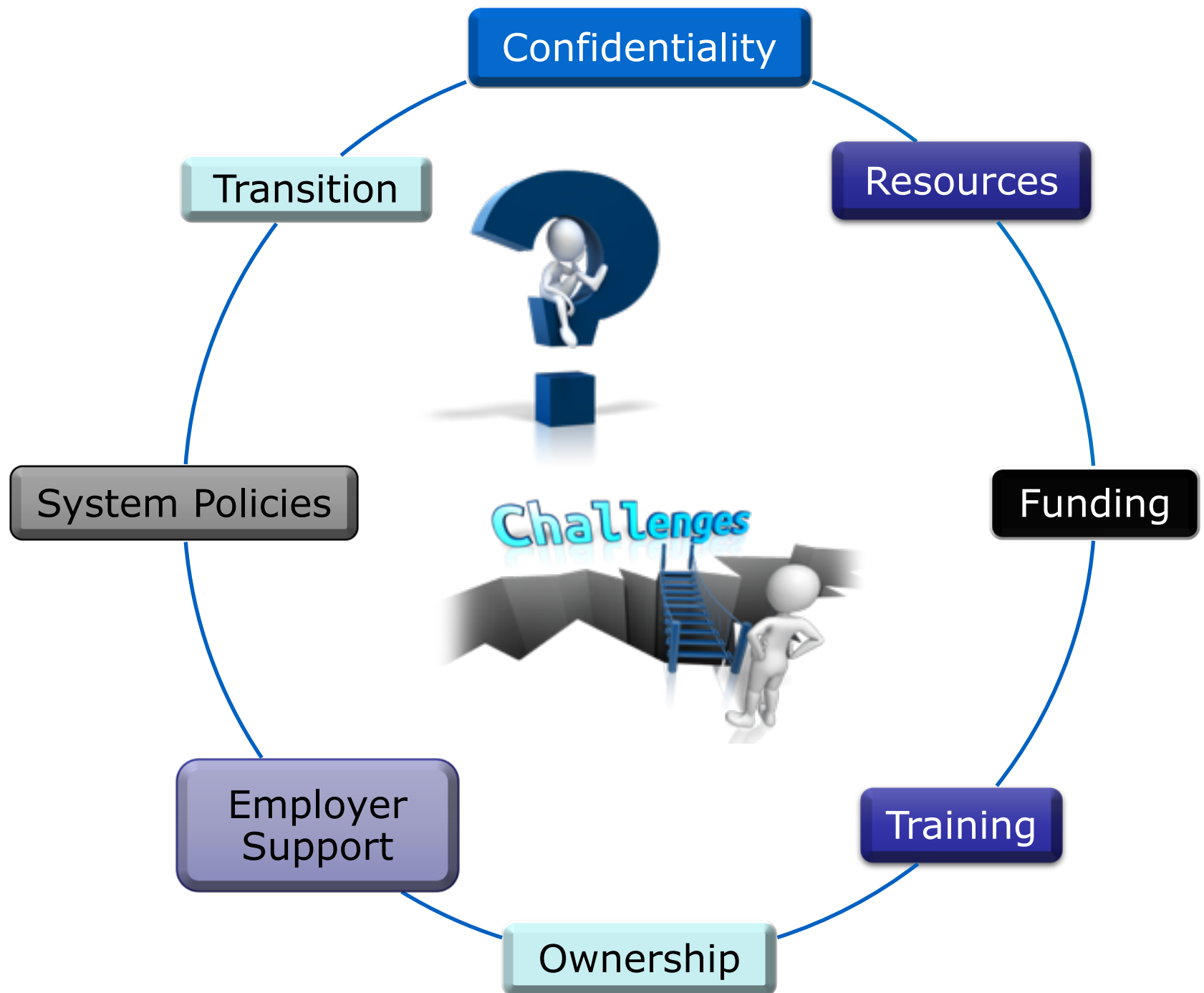
While it is easy to get enamored by the latest app or piece of technology, it is important to use a methodical step-by-step process for deciding what technology works best for a specific need. The following are steps to be done in partnership with the individual:

1. Identify the issue to be addressed.
2. Determine if AT has the potential to address the issue.
3. Review the individual's skills and needs and impact on technology options.
4. Determine the factors in the environment where the technology will be used that need to be considered.
5. Using the information developed, identify potential technology tools.
6. Conduct trial use of the identified technology.
7. Based on this trial, determine if the technology is appropriate and feasible to address the issue identified.
8. Develop a plan to implement the technology, including training the individual and accessing necessary supports.
9. Evaluate the effectiveness of the technology on an ongoing basis.

It is important to view technology as an individualized tool. What works for one person may not be appropriate or helpful for another person.

INSTITUTE FOR COMMUNITY INCLUSION  
University of Massachusetts Boston





# LESSONS LEARNED



- Be creative & willing to try new things
  - Increase expectations & access
    - Return on investment

# Davidisms

“Using technology is 10% knowing the tech and 90% creativity.”

“Everyone can benefit from technology in some way. It is up to us to figure out how.”

**David Baker, MoAT**

# Special Thanks to the Missouri Technology Pilot Learning Community!

- Alece Ausberger, Farmer's House
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- Missy Johns, LOQW
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- Sandy Keyser, MO DMH DD
- Jessica Mahon, ACT
- Chaz Nickolaus, MO DD Council
- Cory Schlegel, ACT
- Duane Shumate, MO DMH DD
- Steven Taylor, MO DMD DD



# THANK YOU

**David Hoff:**

Email: [david.hoff@umb.edu](mailto:david.hoff@umb.edu)

Phone: 617.480.9230

**Jessi Keenoy:**

Email: [jessica.keenoy@umb.edu](mailto:jessica.keenoy@umb.edu)

Phone: 573.303.8770



<https://employmentfirstmo.org>



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